Embedded Controller

# ADC & DAC

**Young-Keun Kim**
**School of Mechanical & Control Engineering**

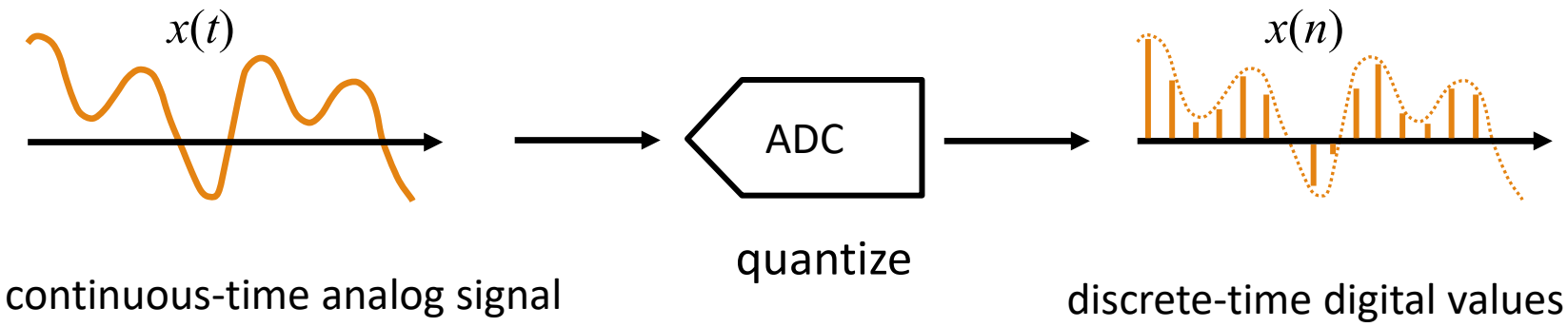
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# ADC (Analog-Digital Converter): AnalogIn

# What is ADC

- **ADC: Analog to Digital Converter (AnalogIn)**
  - Converts a continuous-time voltage signal (analog) within a given range
    - Discrete-time digital values to quantify the voltage's amplitudes
    - Analog signals: Microphone, Strain gauge, Analog voltage and more

  - Three performance parameters:
    - **Resolution**
    - **Sampling rate**
    - **Power dissipation**

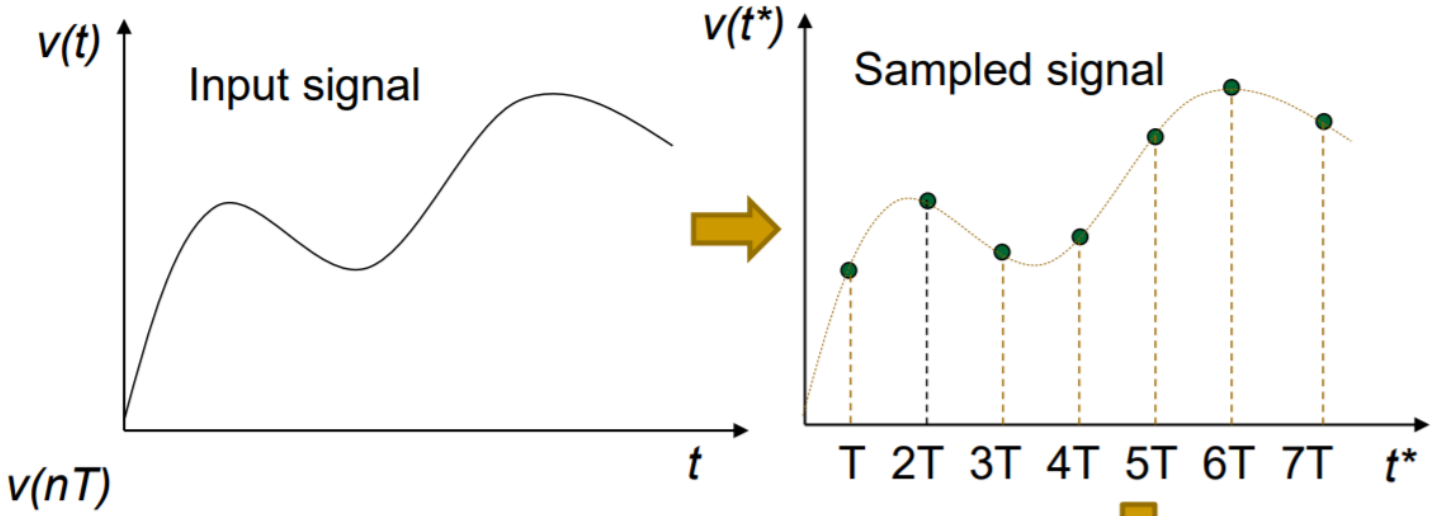




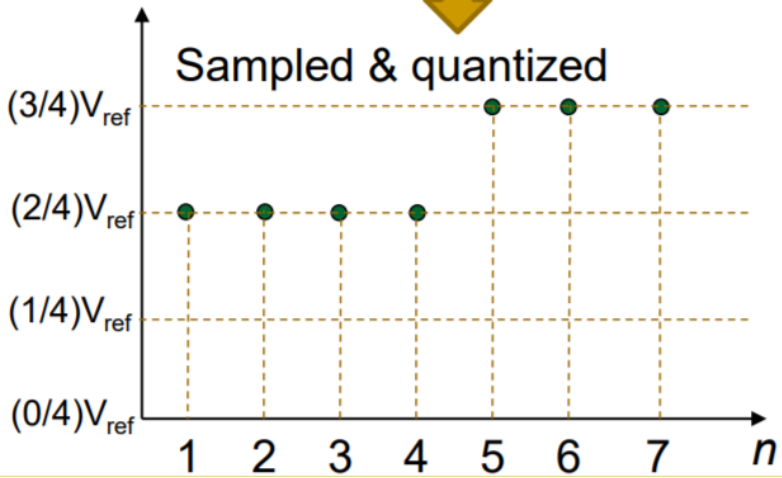

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# What is ADC

- ADC: Analog to Digital Converter




**Ex) 2-bit resolution**




n= 1 2 3 4 5 6 7
b=10 10 10 10 11 11 11 (binary)

v(nT)= (d/4)Vref // d=0~3

[Auburn, Nelsovp]

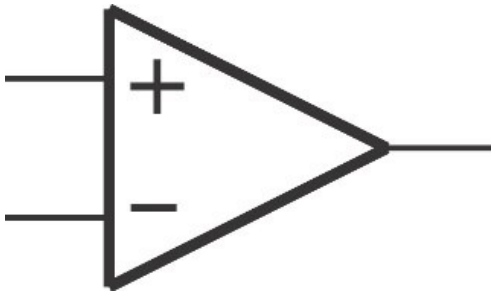


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# ADC Architecture

- **ADC methods:**
  - Common methods
    - Flash
    - Successive-approximation (low powered, used by STM32Fx)
    - Sigma-delta (low-speed)
    - Pipeline (fast speed)

Simplest ADC: Comparator




If ( $V_{in+} > V_{in-}$ ) → $V_{out}$= H

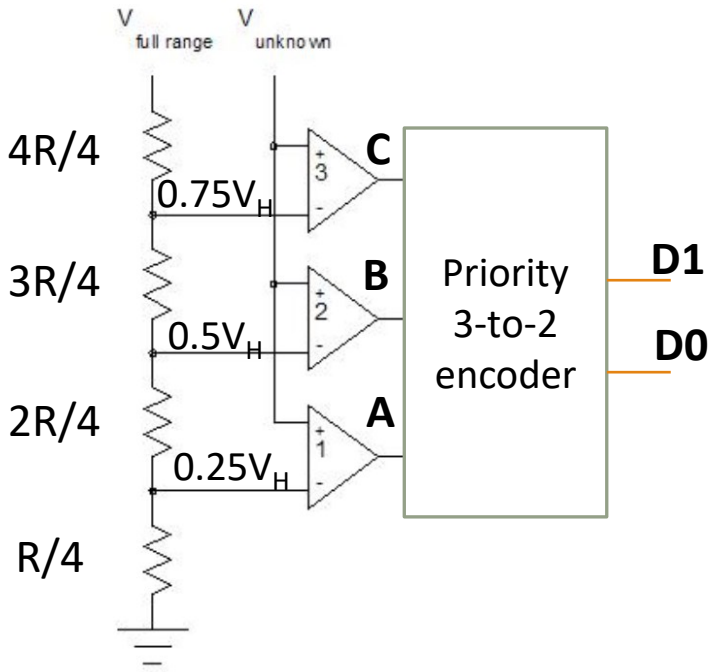


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# ADC Architecture

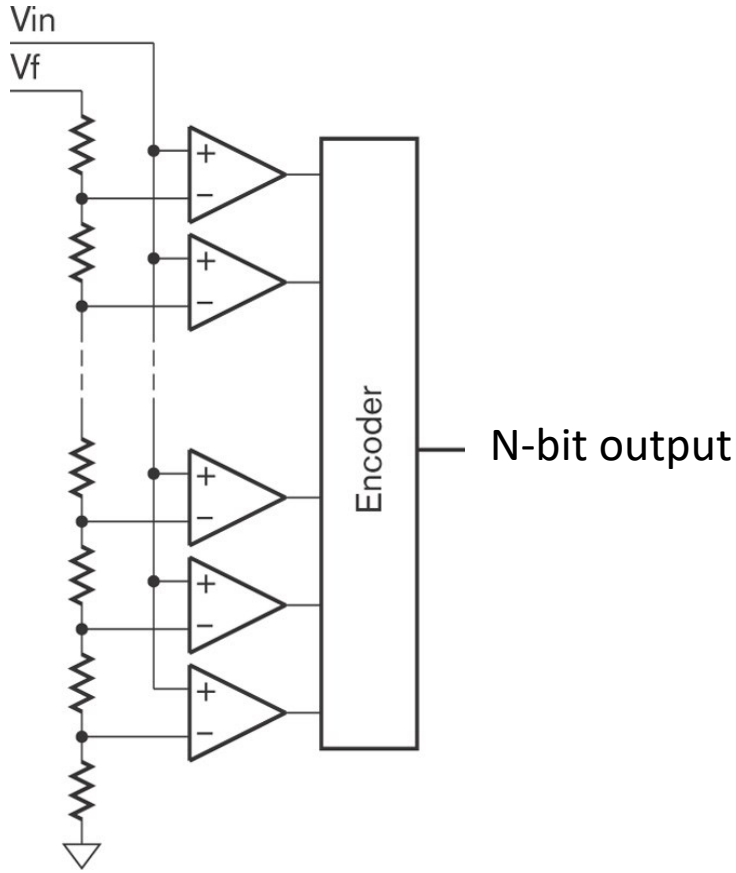
- **Flash ADC**
  - Very fast, but need many comparators
    - N-bit output → $2^N$-1 comparators

(2bit → 3 comparators):




| (input) 3bit | | | (output) 2bit | |
|---|---|---|---|---|
| C | B | A | D1 | D0 |
| 1 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 |

( 3-bit → (8-1)=7 comparators)

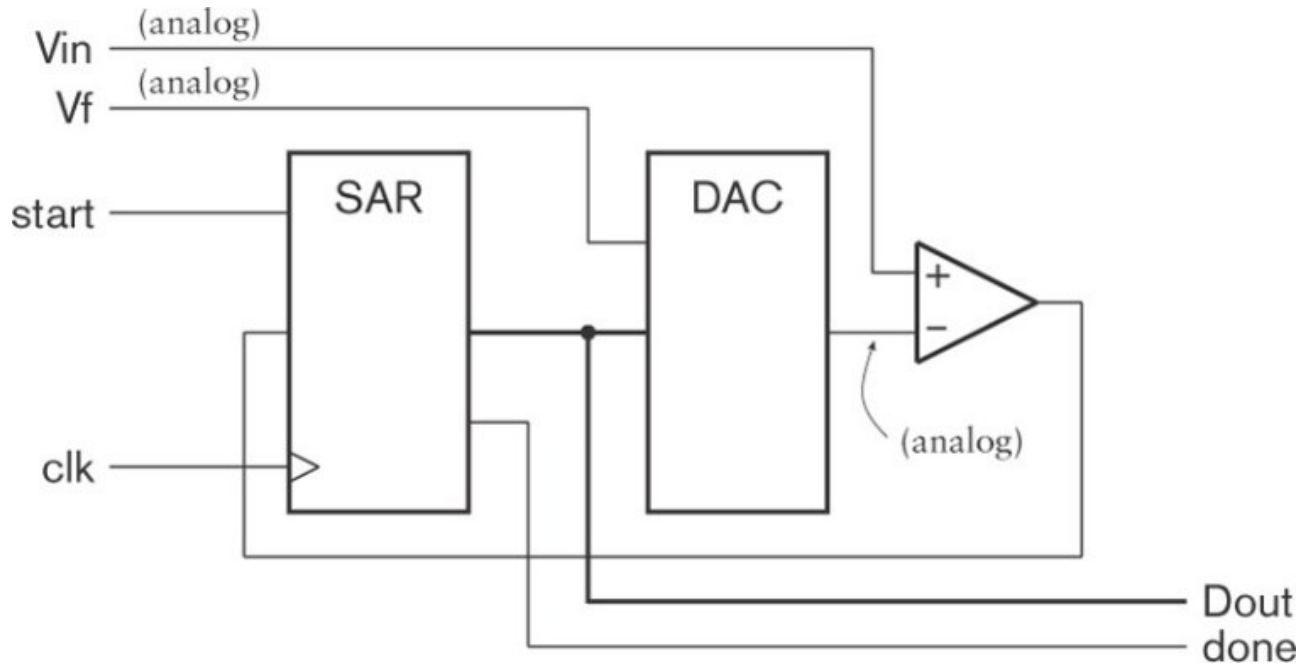





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# ADC Architecture

- **Successive-approximation (SAR) ADC**
  - Slower but fewer circuits: one comparator & DAC
  - About less than 5 MSPS (million samples per sec)
  - Low power dissipation for embedded system

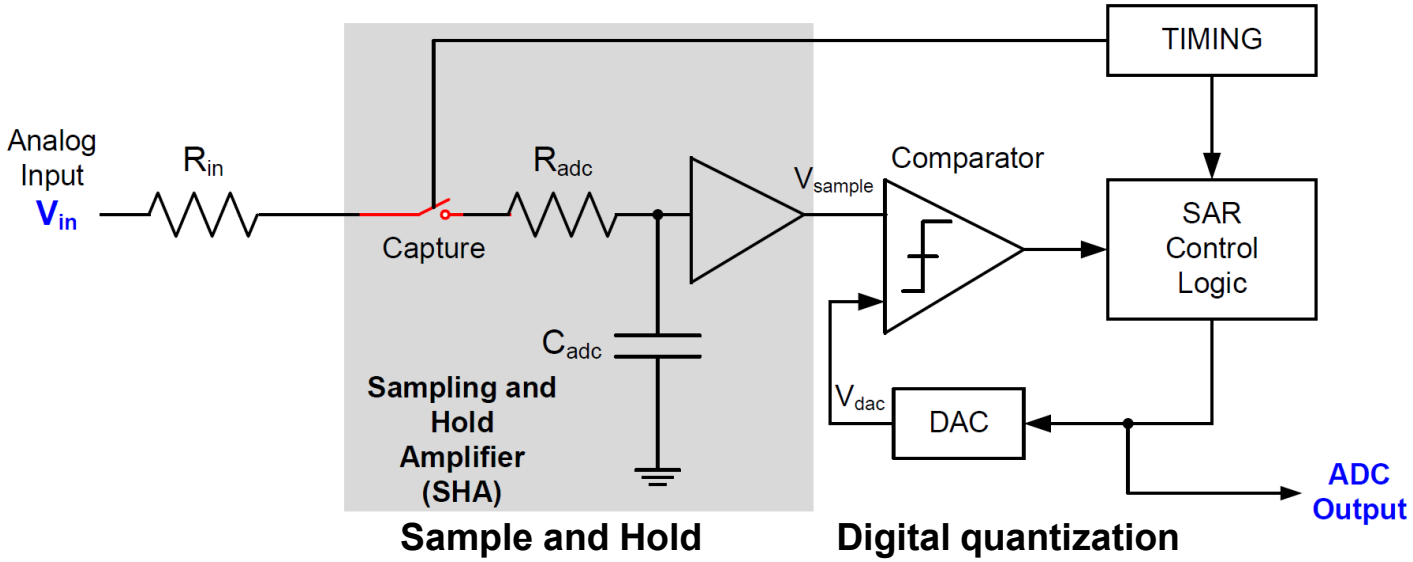


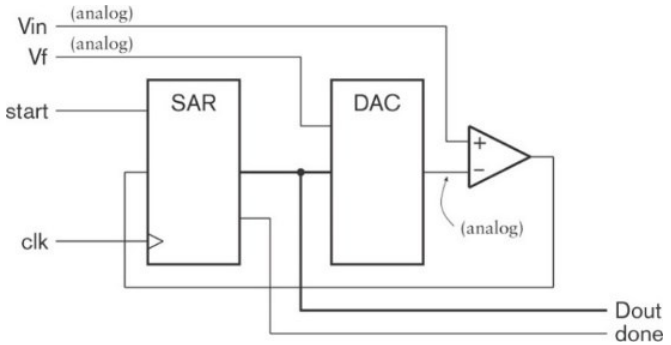



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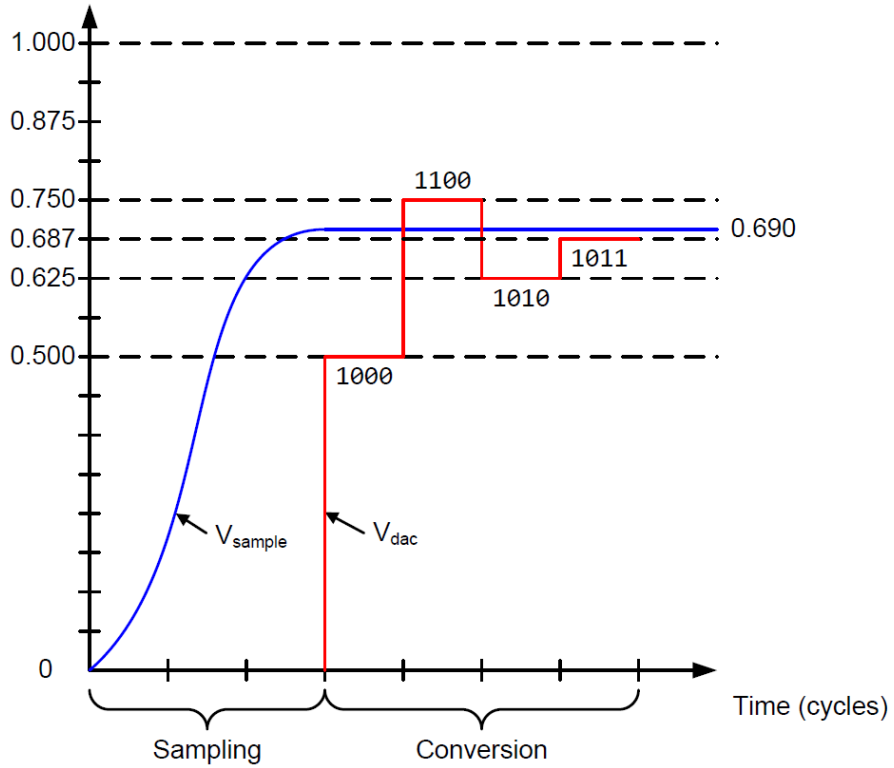
# ADC Architecture

- **Successive-approximation (SAR) ADC**
  - One comparator & DAC




**Ex) Vin=0.69V. 4-bit ADC**



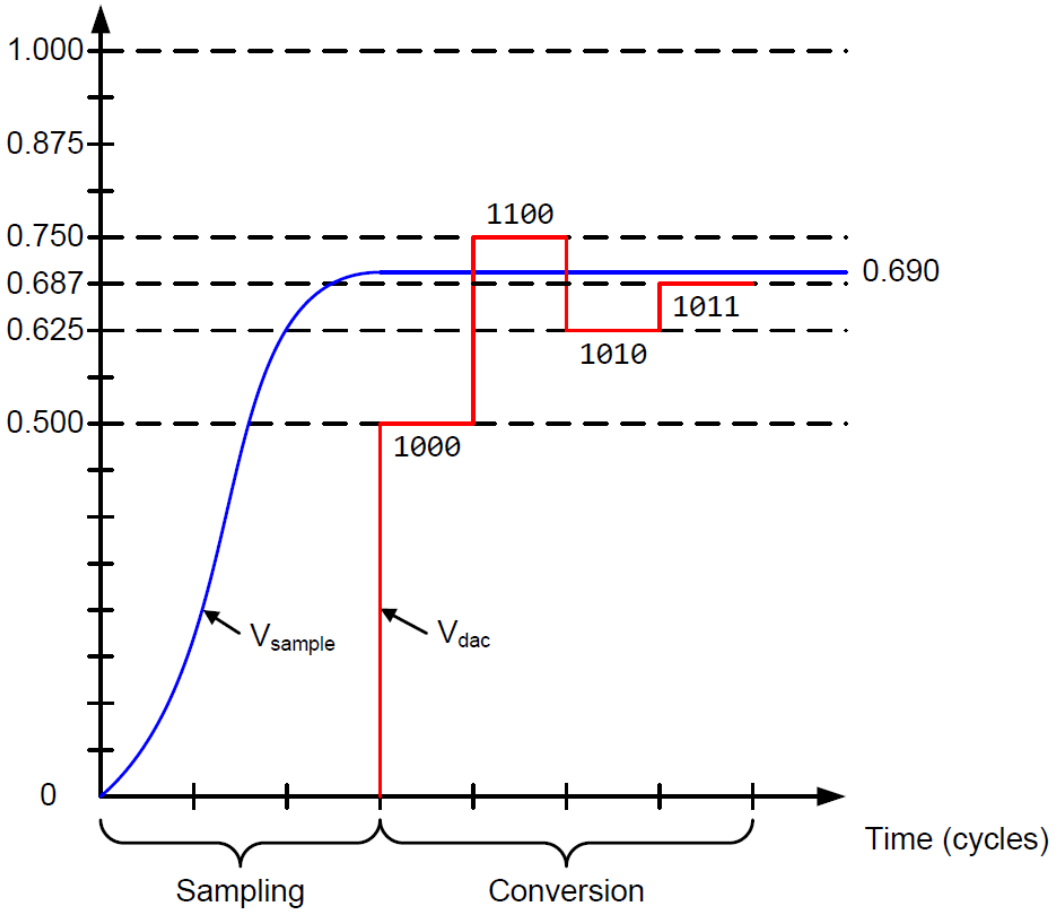



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# ADC Architecture

- **Successive-approximation (SAR) ADC**

**Ex) Vin=0.69V.  4-bit ADC**




- Binary search algorithm to gradually approaches input voltage
- Settle into ±½ LSB bound within the time allowed
- N-bit resolution

$$T_{ADC} = T_{sampling} + T_{Conversion}$$

$T_{sampling}$ = software configurable

$$T_{Conversion} = \mathrm{N} \times T_{ADC_Clock}$$



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- **ADC Conversion Time : SAR ADC**

$$T_{ADC} = T_{sampling} + T_{Conversion}$$

Ex) ADC_CLK = 16 MHz, Sampling time = 4 cycles

For 12-bit ADC: $T_{ADC} = 4 + 12 = 16\ cycles \rightarrow 1\mu s$ / sample (1MSPS)

For 6-bit ADC: $T_{ADC} = 4 + 6 = 10\ cycles \rightarrow 625ns/sample$

The conversion sampling should be faster than Nyquist rate
- Nyquist rate ≥ 2 x (highest frequency bandwidth)

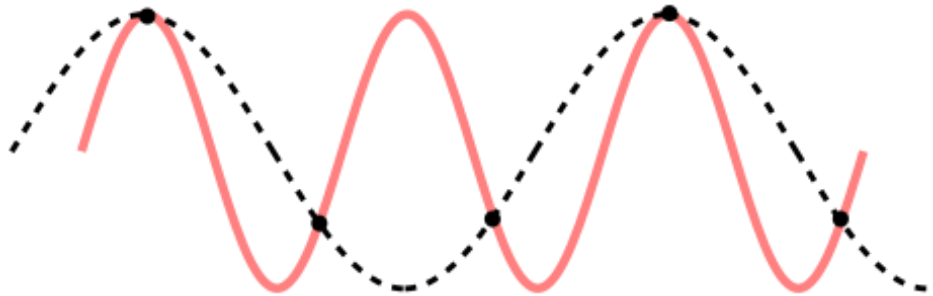


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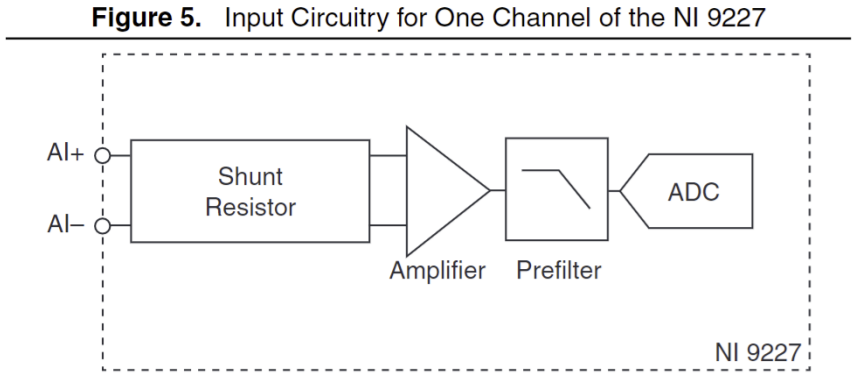
# Minimum Sampling Rate:

- **Minimum Sampling Rate: Nyquist–Shannon Sampling Theorem**
  - To reconstruct the analog input signal
    - Sampling rate should be **at least twice the maximum frequency component** contained in the input signal

Ex) aliasing example



[wiki]




Antialiasing filter:

Pre-filtering: Analog circuit to filter out high-frequency components

Post-filtering: Oversample continuous signal → Filter out high-frequency components by averaging



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# Resolution(Precision)

**Resolution** is determined by number of bits (in binary) to represent an analog input

- STM32F can select 6~12-bits ADC.

Max quantization error: $\Delta = V_{REF}/(2^N - 1)$

- For 5V, 3-bit resolution: $5V/(2^3-1)=$ 710 m V / bit
- For 5V, 12-bit resolution: $5V/(2^{12}-1)=$ 1.22 mV / bit

N=3, $V_H$=5, $V_L$=0, $V_{in}$=4

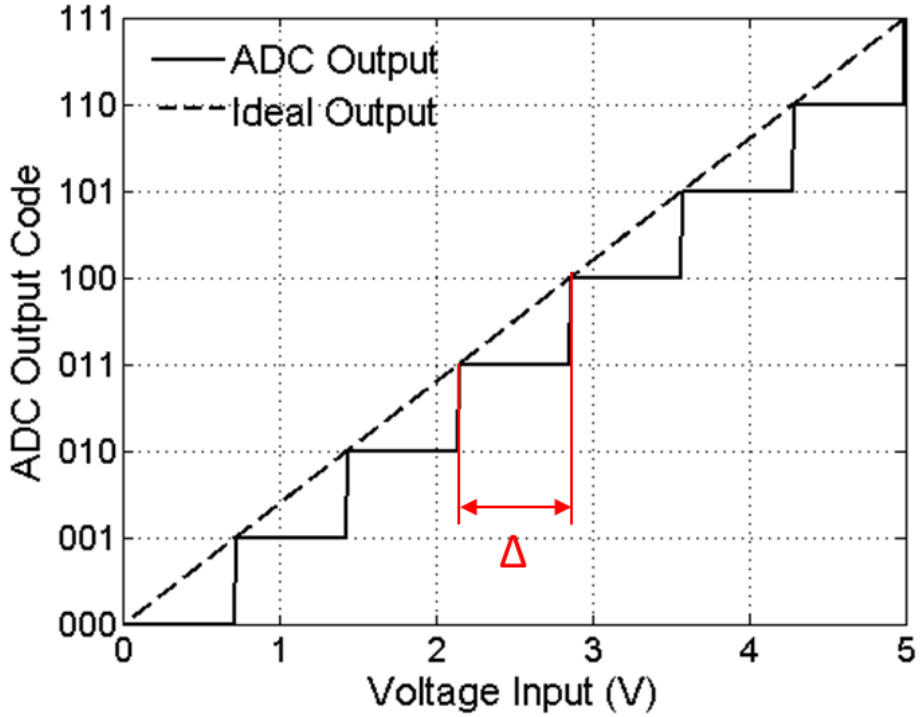
1) Max quantization error:

$\Delta = V_{REF}/(2^N - 1) = 5/(2^3 - 1) = 5/7 = 0.71$ [V/code]

2) Quantized output:

$$OUT = \text{floor}\left((2^N - 1)\frac{V_{IN} - V_L}{V_H - V_L}\right)$$

OUT= $(2^3 - 1)(4/5)$=round(5.25) = 5 = b101






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# Reference Voltage

- $V_{REF}$ can be positive and negative within the allowable range
- Check the development board for $V_{REF}$ default value. (max.3.6V)

**Table 39. ADC pins**

| Name | Signal type | Remarks |
|---|---|---|
| $V_{REF+}$ | Input, analog reference positive | The higher/positive reference voltage for the ADC, 1.8 V ≤$V_{REF+}$ ≤$V_{DDA}$ |
| $V_{DDA}$ | Input, analog supply | Analog power supply equal to $V_{DD}$ and 2.4 V ≤$V_{DDA}$ ≤$V_{DD}$ (3.6 V) for full speed 1.8 V ≤$V_{DDA}$ ≤$V_{DD}$ (3.6 V) for reduced speed |
| $V_{REF-}$ | Input, analog reference negative | The lower/negative reference voltage for the ADC, $V_{REF-}$ = $V_{SSA}$ |
| $V_{SSA}$ | Input, analog supply ground | Ground for analog power supply equal to $V_{SS}$ |
| ADCx_IN[15:0] | Analog input signals | 16 analog input channels |

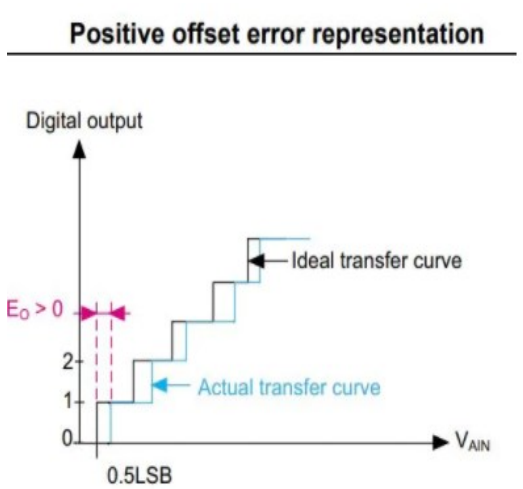


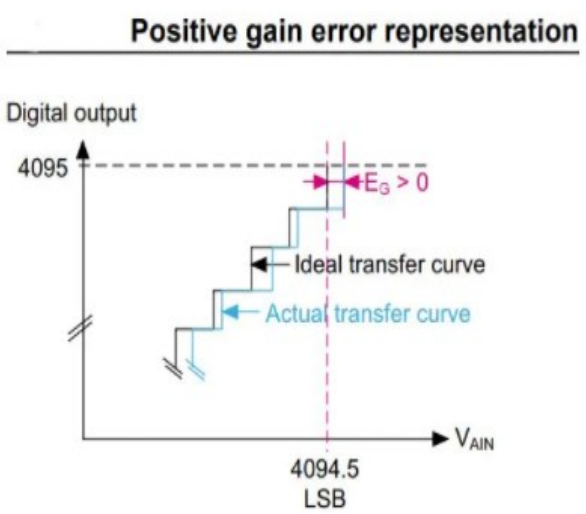
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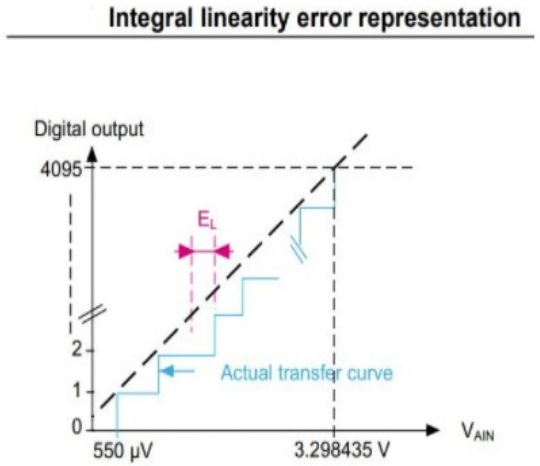
# ADC errors

## ADC Errors

- Offset Error: (First actual transition – ideal transition).  Can be calibrated
- Gain Error:  (Last actual transition – ideal transition)
- Integral Linearity Error: maximum deviation between actual transition and the endpoint correlation line
- ADC Reference Voltage noise  (Vref, $V_{DDA}$)
- Analog Input Signal Noise
- Dynamic Range Bad Matching
- And others
  - Signal source capacitance, voltage drop by impedance, IO pin cross-talking
  - EMI-induced noise




https://deepbluembedded.com/stm32-adc-tutorial-complete-guide-with-examples/



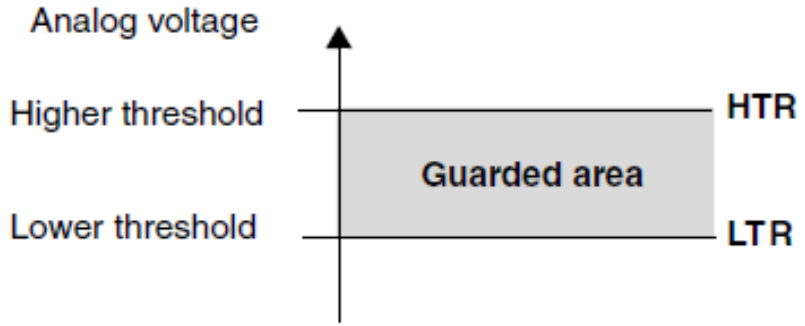
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# ADC in STM32

- **ADC in MCU (STM 32f4x)**
  - 12-bit ADC:  successive approx. ADC
  - 19 MUX channel:  16 external source, 2 internal, Vbat

  - **Conversion mode:**
    - Single / Continuous conversion mode
  - **Scan mode:**
    - Single channel  / a Group of channels
    - Order of channels can be configured
  - **Group:**
    - Regular group
    - Injected group: ~4 channel groups that have higher priority

  - **Interrupt:**
    - End of Conversion interrupt ( EOC )
    - Analog watchdog (AWD): if Input voltage is out of range






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# ADC in STM32

**ADC Configuration**

- ADC_CLK, Sample Time, Resolution → Total conversion time= Sample Time + Conversion
- Group: Regular or Injected
- GPIO Pin: Input pin for ADC
- Repetition: Single Conversion or Continuous Conversion
- Sequence: One channel or Multiple channel

**Start ADC Conversion**

- Software Trigger
- External Trigger: EXTI / Timer

**Read ADC Conversion**

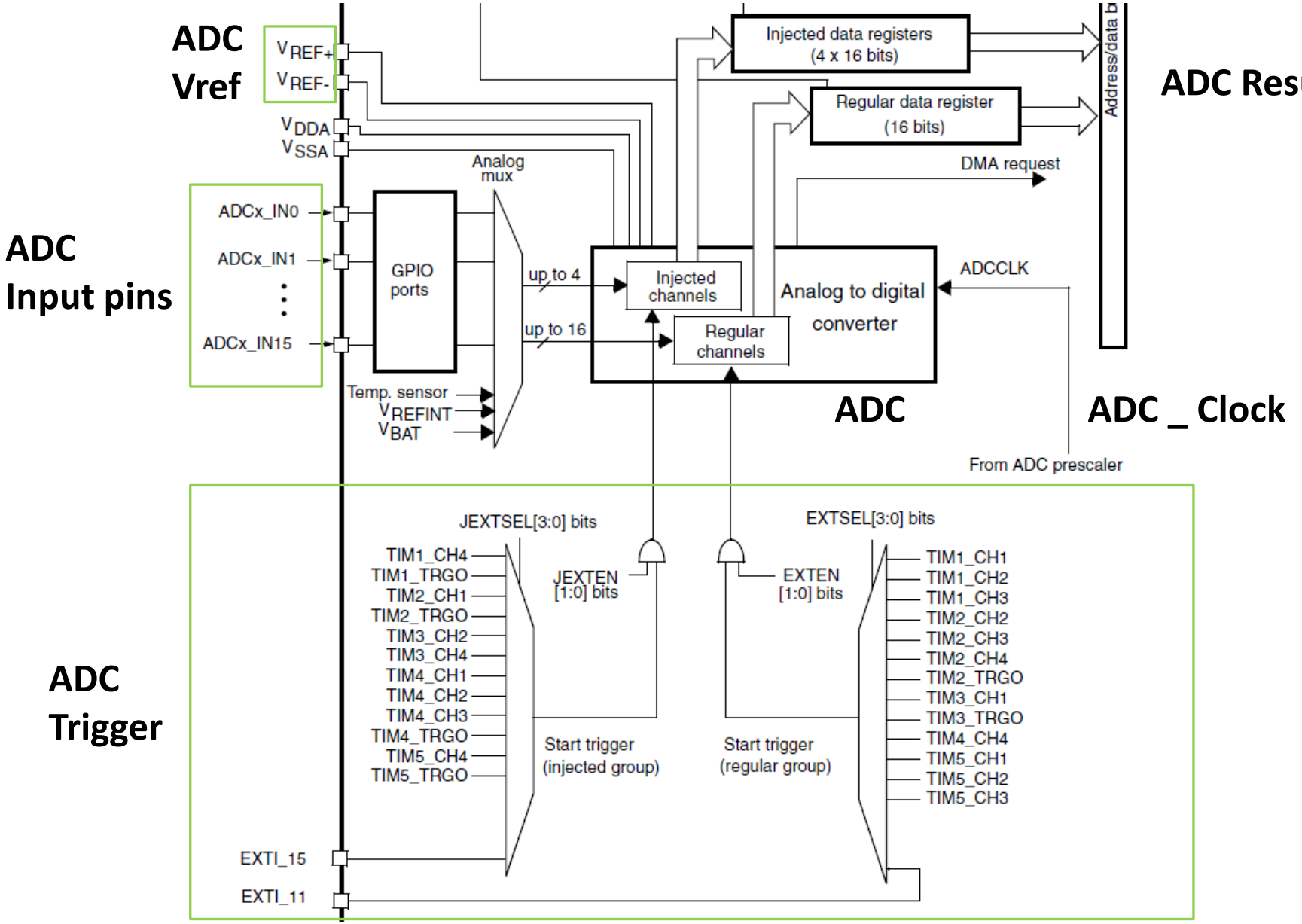
- Polling
- Interrupt: EOC
- DMA*



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# ADC in STM32

- ADC



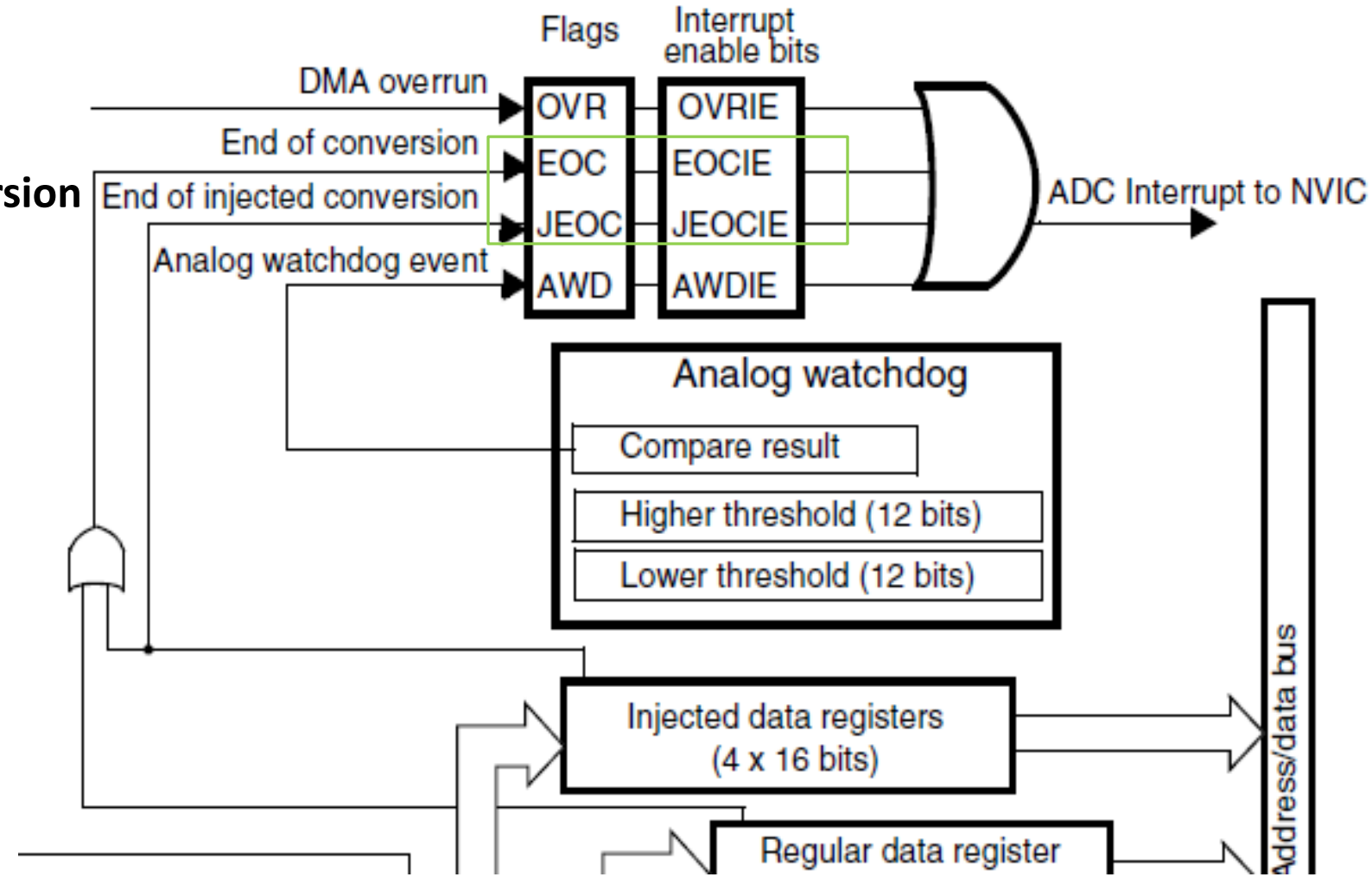



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# ADC in STM32

- Interrupt from ADC

(J)EOC: End of Conversion






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# ADC Mode

## Mode Configurations

| | Options | |
|---|---|---|
| **Group** | Regular group | Injected Group |
| **Repetition** | Single Conversion | Continuous Conversion |
| **# of Channel** | One Channel | Scan Conversion(Multi-Channel) |

## Others

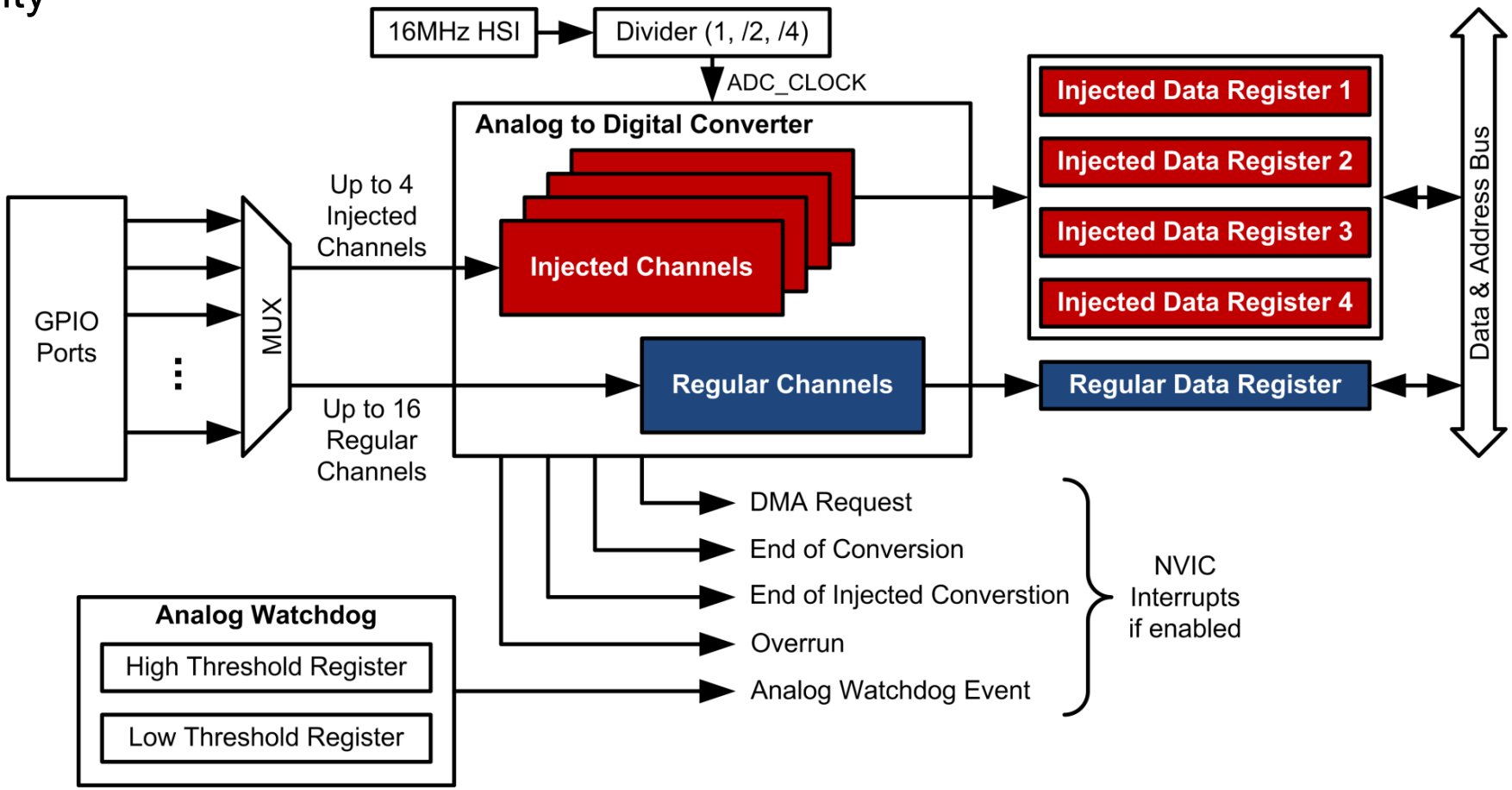
- Interleaved mode
  - Multiple ADC blocks per one ADC input pin
  - While one ADC block is processing conversion, other ADC block samples the same pin
  - Can increase Sampling Per Second (SPS)



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# Group: Regular, Injected

- **Regular Group**
  - Up to 16 channels,  only **one shard Data Register**
- **Injected Group**
  - Up to 4 channels, each channel has  **its own Data Register**
  - Higher priority





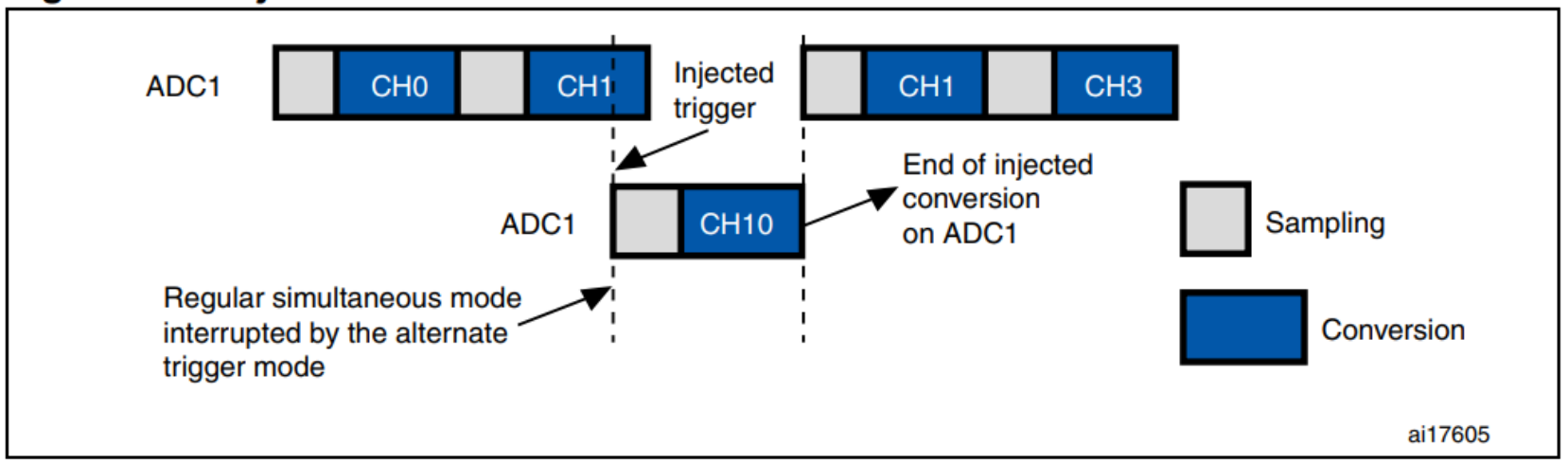

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# Group: Regular, Injected

- **Injected Groups vs Regular Groups**
  - Injected Group: Can interrupt conversion of current channel in regular channel group

## Injected Conversion Mode:

**Figure 6. Injected conversion mode**



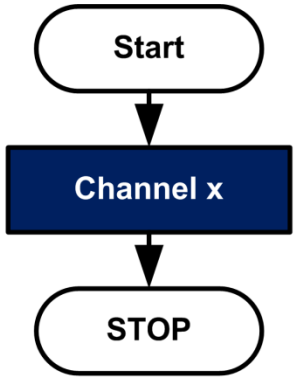



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# ADC Mode

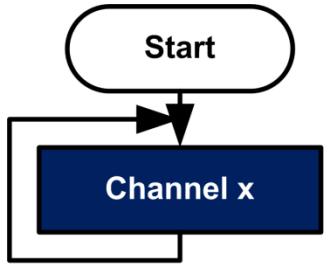
- **Single-Channel Mode** vs **Scan Mode**(multi-channels)
  - Can choose the order of channels

- **Single-Conversion Mode vs Continuous-Conversion Mode**
  * Discontinuous mode:  Can choose a part of channels to be different triggered



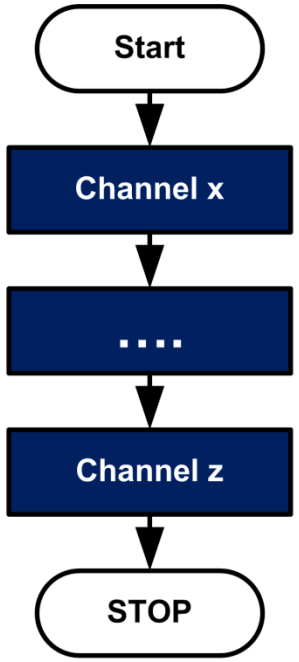

**Single Channel, Single Conversion Mode**

*CONT in ADC_CR2 = 0*
*SCAN in ADC_CR2 = 0*



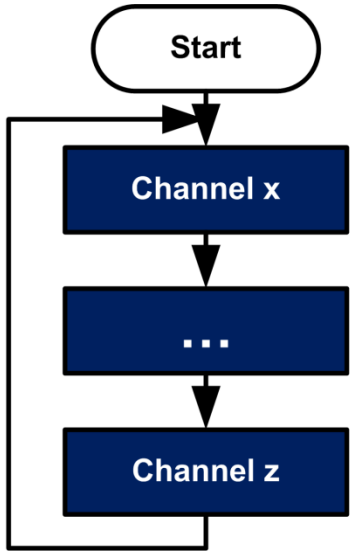

**Single Channel, Continuous Conversion Mode**

*CONT in ADC_CR2 = 1*
*SCAN in ADC_CR2 = 0*




**Scan Mode with Single Conversion**

*SCAN in ADC_CR2 = 1*
*CONT in ADC_CR2 = 0*




**Scan Mode with Continuous Conversion**

SCAN in ADC_CR2 = 1
*CONT in ADC_CR2 = 1*



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# Reading ADC values

## Reading ADC

- Polling Method
  - Not efficient, but the easiest.
  - Start the conversion and wait until it completes the conversion

- **Interrupt Method**
  - More efficient.
  - But, when multiple channels are in circular mode, you can have periodic interrupts too much burden for CPU.
  - Can cause latency and other timing issues
  - It can be avoided by using DMA

- *DMA method
  - Most efficient way to convert multiple ADC at very high rate without CPU intervention



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# Triggering ADC

- **How to start ADC process?**
  - Software Trigger:  User Input Command
  - Hardware Trigger:  Timer, EXTI etc

- **Hardware Trigger**
  - External Trigger : External input
  - External Trigger : Timer

- **Software Trigger**
  - HAL_API:  ADC_start()



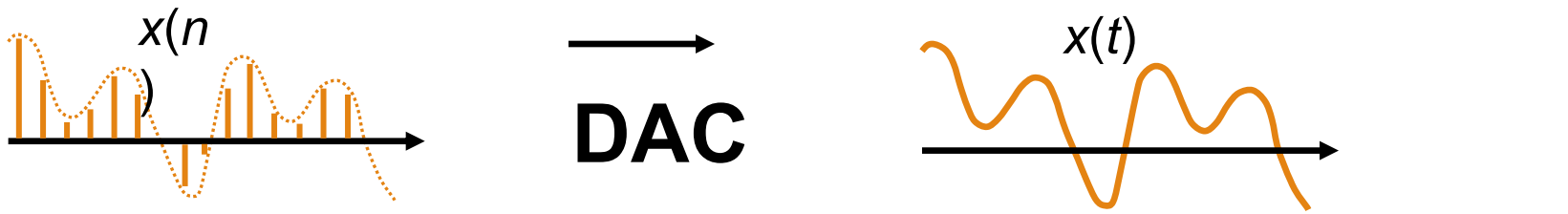
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# DAC (Digital-Analog Converter): AnalogOut

# What is DAC

- **DAC: Digital to Analog Converter**
  - Converts digital data into a voltage (analog) signal by a N-bit DAC
  - There is no DAC in STM32f411.
    - You can externally connect to a DAC chip via I2C, SPI etc.
  - Application examples:
    - digital audio, waveform generation
  - Three performance parameters:
    - resolution
    - sampling rate
    - power dissipation

$$DAC_{output} = V_{ref} \times \frac{(Digital\ Value)}{2^N - 1}$$




discrete-time digital values

continuous-time analog signal



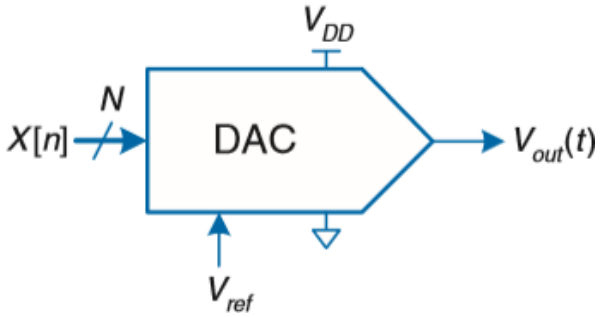
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# What is DAC

- **DAC**
  - Analog filter could be used to smooth out the step-like outputs
  - Characterized by their resolution, dynamic range, sampling rate, and accuracy

  - Resolution
    - How many bits used to represent data. N-10 bits etc
  - Dynamic Range
    - Range of Vref- to Vref+ (0~5V)
  - Sampling rate
    - Samples per second, e.g. Fs=44kHz
  - Accuracy
    - Example: +-3 least significant bits (LSB)

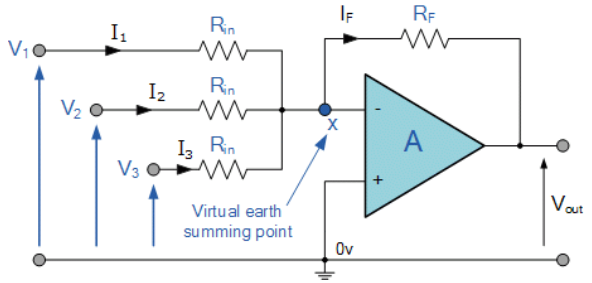
$$V_{out}(t) = \frac{X[n]}{2^N} V_{ref}$$



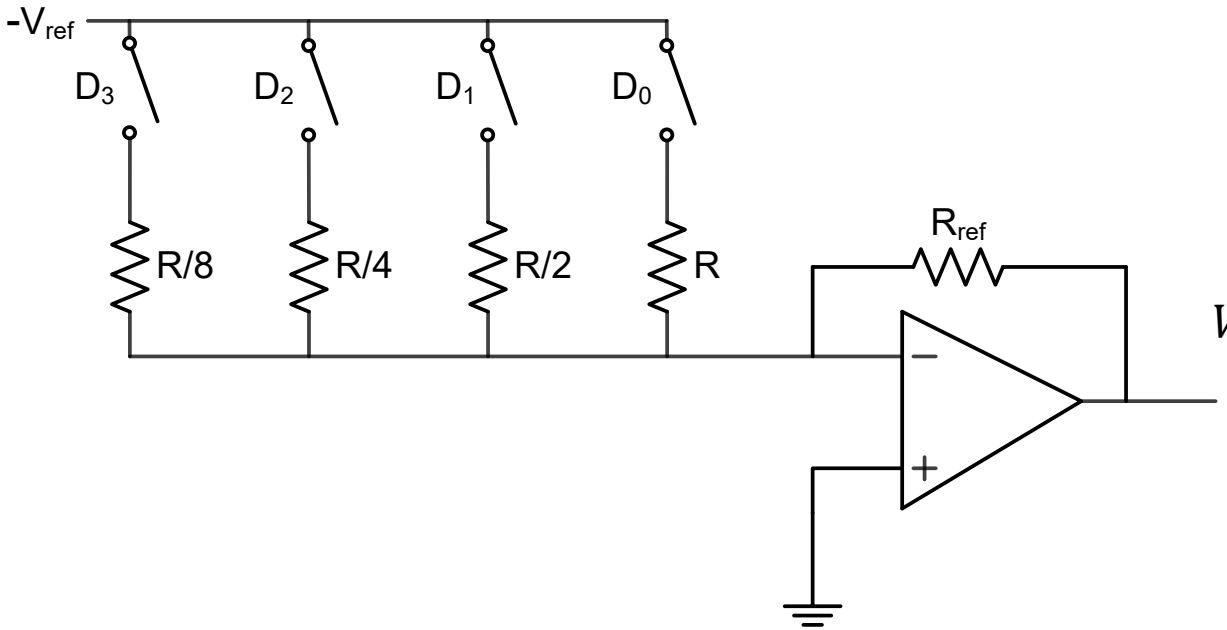
# DAC methods

- **DAC methods**
  - Binary-weighted resistor
  - R-2R ladder (A special case of binary-weighted resistor)
  - Pulse-width modulator (PWM)

OP-AMP adder




**Binary-weighted resistor**




$$V_{out} = V_{ref} \times \frac{R_{ref}}{R} \times (D_3 \times 2^3 + D_2 \times 2^2 + D_1 \times 2 + D_0)$$

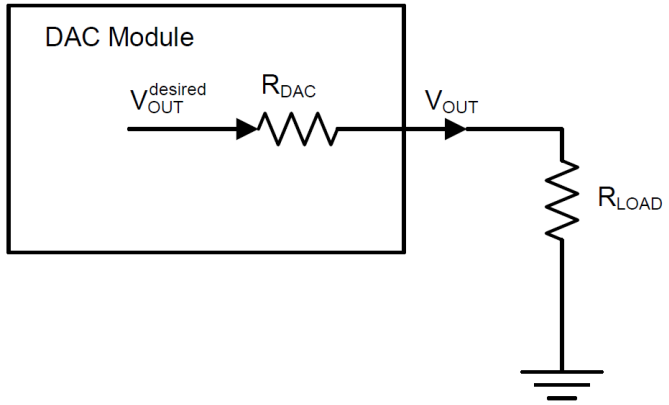


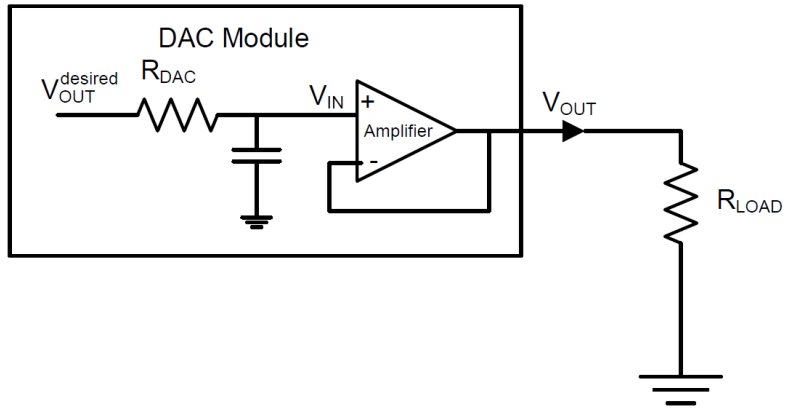
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# Buffered Output

## Load Effect

- Load resistor must be high enough compared to internal resistor:  causes load effect
- Use a buffer to remove  load effect




$$V_{OUT} = \frac{R_{LOAD}}{R_{DAC} + R_{LOAD}} \times V_{OUT}^{desired}$$

$$V_{OUT} \approx V_{OUT}^{desired}$$

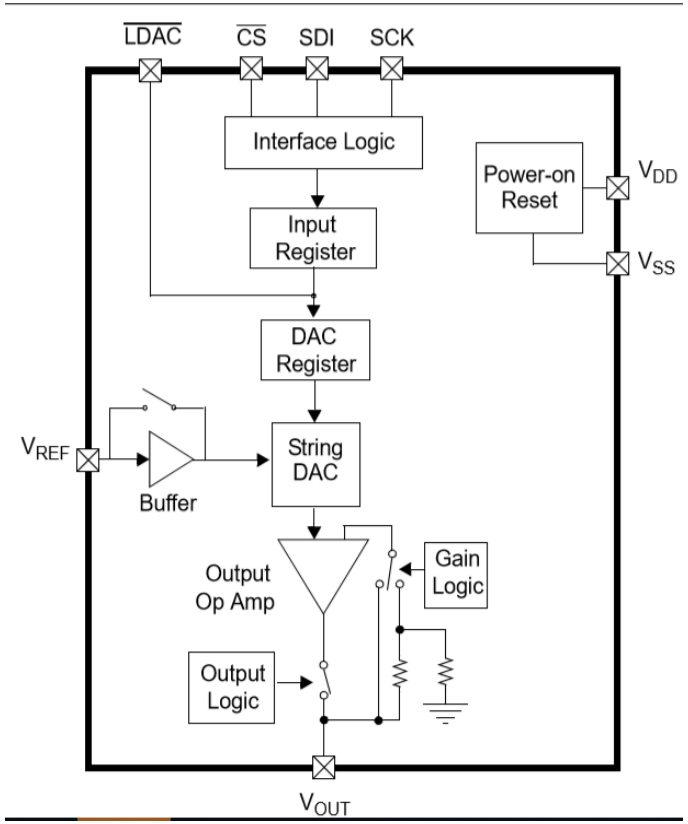


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# DAC in STM32F4

- **There is no DAC in Nucleo32**
  - Other models of Cortex M4 has DAC embedded.
  - Also, you can connect to DAC modules via SPI, I2C protocols
  - Read more about DAC in other reference textbook

**Case Study: MCP 4911 DAC**




MICROCHIP **MCP4901/4911/4921**

**8/10/12-Bit Voltage Output Digital-to-Analog Converter with SPI Interface**

**Features**

- MCP4901: 8-Bit Voltage Output DAC
- MCP4911: 10-Bit Voltage Output DAC
- MCP4921: 12-Bit Voltage Output DAC
- Rail-to-Rail Output
- SPI Interface with 20 MHz Clock Support
- Simultaneous Latching of the DAC Output with $\overline{\text{LDAC}}$ Pin
- Fast Settling Time of 4.5 µs
- Selectable Unity or 2x Gain Output
- External Voltage Reference Input
- External Multiplier Mode
- 2.7V to 5.5V Single-Supply Operation
- Extended Temperature Range: -40°C to +125°C



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